

5 People with cystic fibrosis produce very thick, sticky mucus.

Cystic fibrosis is caused by mutations in a gene coding for the CFTR protein.

(a) Explain why a mutation in this gene results in the production of very thick, sticky mucus.

(3)

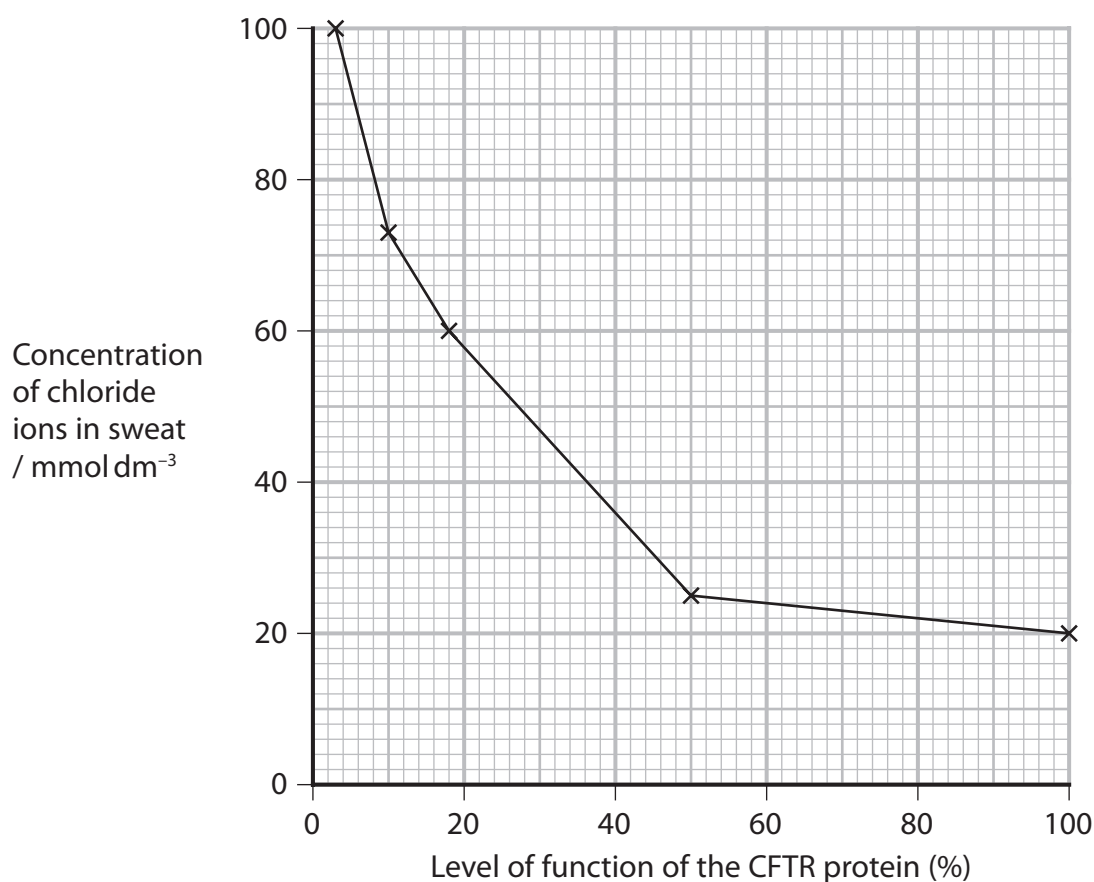
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- (c) The graph shows the correlation between the concentration of chloride ions in sweat and the level of function of the CFTR protein.



Individuals diagnosed with cystic fibrosis have a level of function of the CFTR protein of 18% or less.

- (i) Which is the change in concentration of chloride ions in the sweat of an individual when the level of function of CFTR protein decreased from 100% to 18%?

(1)

- ☐ **A** 15 mmol dm^{-3}
- ☐ **B** 35 mmol dm^{-3}
- ☐ **C** 40 mmol dm^{-3}
- ☐ **D** 80 mmol dm^{-3}



(ii) Cystic fibrosis results from different mutations in the CFTR gene.

Explain how the graph provides evidence that cystic fibrosis results from different mutations in the CFTR gene.

(2)

(Total for Question 5 = 11 marks)

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